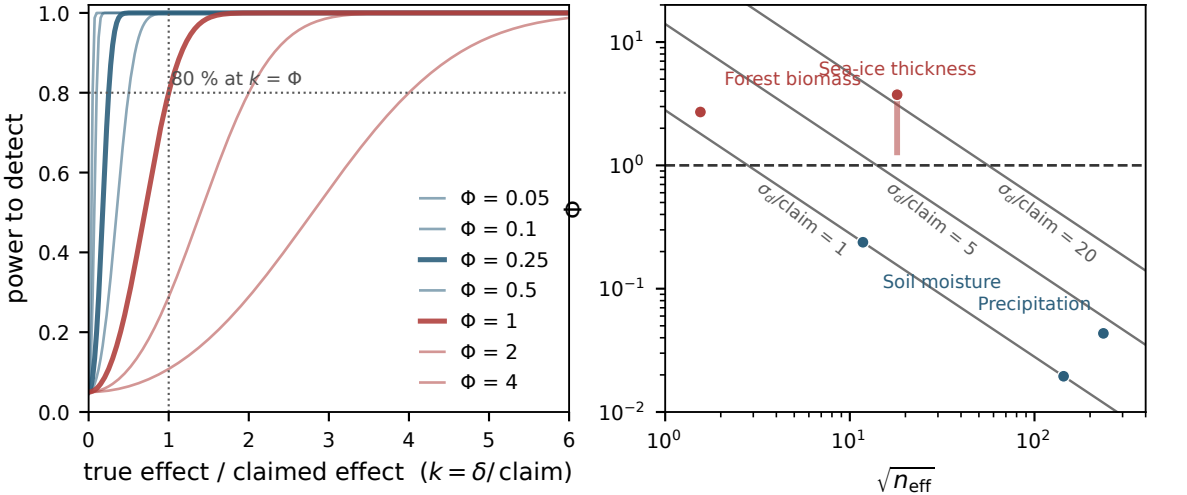
(a) the criterion is a power statement
power to detect
true effect / claimed effect ($k = \delta/\text{claim}$)
80 % at $k = \Phi$
Φ = 0.05
Φ = 0.1
Φ = 0.25
Φ = 0.5
Φ = 1
Φ = 2
Φ = 4
(b) Φ falls as independent information grows
Forest biomass
Sea-ice thickness
Soil moisture
Precipitation
σ_δ/claim = 1
σ_δ/claim = 5
σ_δ/claim = 20
$\sqrt{n_{\text{eff}}}$